

Seasonal adjustment of economic data

Higher frequency time series: DSA and other methods

Daniel Ollech, Lea Hengen / Deutsche Bundesbank, DG Statistics
Seasonal adjustment, November 7, 2025

Agenda

- Motivation
- `dsa()` Introduction
- `dsa()` with calendar regressors
- Alternative seasonal adjustment approaches
- Key notions of higher frequency time series
- Summary

I Motivation (I/VI)

- Covid-19-pandemic: Need for more timely and granular information on the state of the economy
- Increased use of daily and weekly time series
- How to seasonally adjust these series?
- What are import features of higher frequency time series?

Motivation (II/VI)

Lewis, Mertens, Stock (2020)

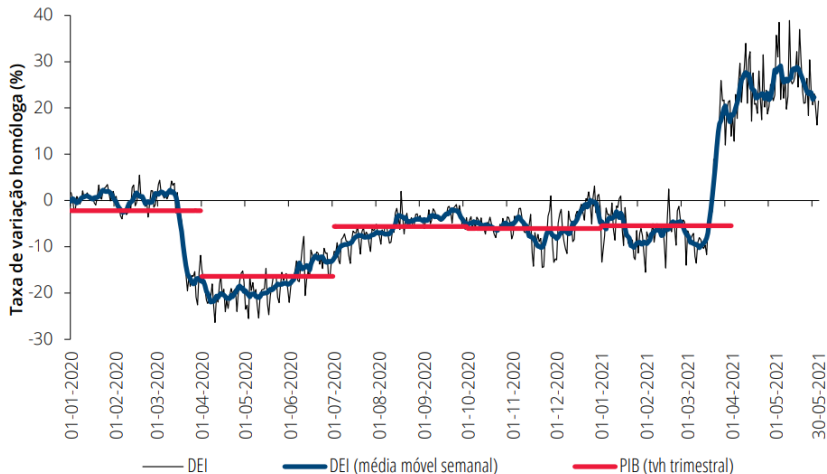


- Weekly composite indicator of:
 - Redbook same-store sales
 - Rasmussen Consumer Index
 - Claims for unemployment insurance (new, continued)
 - Income/employment tax withholdings
 - Railroad traffic
 - American Staffing Association Staffing Index
 - Steel production
 - Wholesale sales of gasoline, diesel, and jet fuel
 - Electricity load
- “Seasonal adjustment”: year-on-year growth rate

Motivation (IV/VI)

Lourenço and Rua (2021)

Gráfico 1



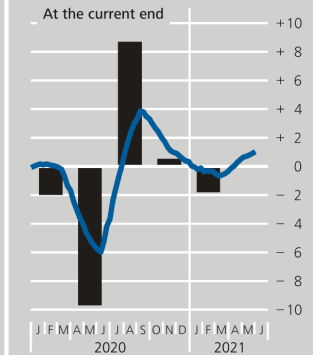
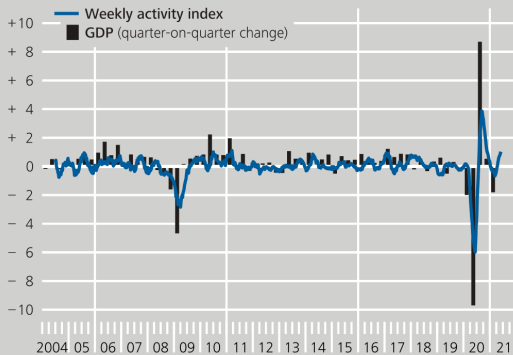
- Daily composite indicator of:
 - Road traffic of heavy commercial vehicles
 - Electricity consumption
 - Natural gas consumption
 - Cargo and mail landed
 - Card-based payments
- “Seasonal adjustment”: STL for weekday effects + RegARIMA + year-on-year growth rate

Motivation (V/VI)

Eraslan and Goetz (2020)

Weekly activity index and GDP growth

%



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- Weekly composite indicator of daily and weekly series:
 - Electricity consumption
 - Daily truck toll mileage
 - Worldwide number of flights
 - Pedestrian frequency
 - Air pollution
 - ICC: Consumer confidence
 - Google Trends: Unemployment, short time work, state support
 - Credit card payments
- Seasonal adjustment: DSA and WSA

Exercise

In groups of 3 look at daily time series

- What are unique characteristics of daily time series compared to monthly/quarterly series?

Group 1: `dsa::daily_data$currency_circulation`

Group 2: `dsa::daily_data$no2` (Air quality)

Group 3: `dsa::daily_data$elec_consumption` (Electricity)

Group 4: `mex_currency_circulation`

Group 5: `kor_electricity`

Group 6: `nz_departures`

**Hint: Use `library(xts)`*

■ Daily data, basics (I/III)

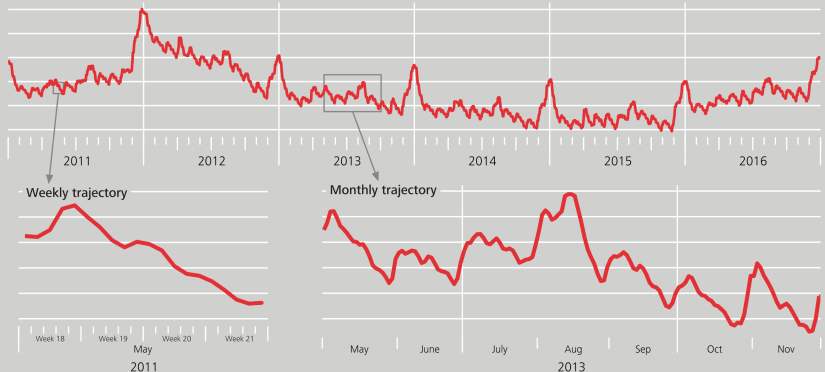
Discussion

How are daily data different from monthly data?

Daily data, basics (II/III)

Currency in Circulation I

€ bn

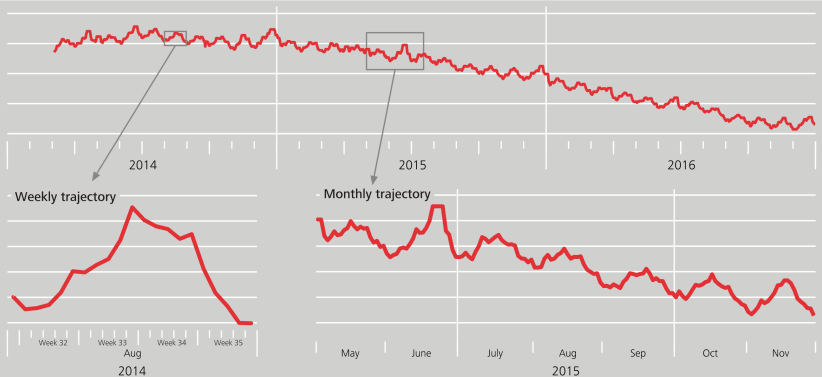


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Daily data, basics (III/III)

Currency in Circulation II

€ bn



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Exercise

What are examples for...

- Weekly seasonality?
- Monthly seasonality?
- Annual seasonality?
- Calendar effects?

Sources of seasonality (I/II)

Currency in circulation: dramatically simplified economic rationale

People...

- need money for the weekend $\rightsquigarrow$ **Weekly** spike
- get paid at the end of the month $\rightsquigarrow$ **Monthly** spike
- need money for Christmas presents $\rightsquigarrow$ **Annual** spike
- need money for holidays $\rightsquigarrow$ Spike around **Easter**

Sources of seasonality (II/II)

Currency in circulation: dramatically simplified economic rationale

People...

- need money for the weekend $\rightsquigarrow$ Weekly spike
 - get paid at the end of the month $\rightsquigarrow$ Monthly spike
 - need money for Christmas presents $\rightsquigarrow$ Annual spike
 - need money for holidays $\rightsquigarrow$ Spike around Easter
- } Seasonal
- } Calendar

Unobserved component model

Monthly data

$$Y_t = T_t + S_t + C_t + I_t$$

Daily data (without cross-seasonal effects)

$$Y_t = T_t + S_t^{(7)} + S_t^{(31)} + S_t^{(365)} + C_t + I_t$$

Unobserved component model

Monthly data

$$Y_t = T_t + S_t + C_t + I_t$$

Daily data (without cross-seasonal effects)

$$Y_t = T_t + S_t^{(7)} + S_t^{(31)} + S_t^{(365)} + C_t + I_t$$

Daily data (example with cross-seasonal effects)

$$Y_t = T_t + S_t^{(7)} + S_t^{(365)} + S_t^{(7 \times 365)} + C_t + I_t$$

■ Seasonal adjustment of daily time series: DSA

- Step I: Adjust intra-weekly seasonality with **STL**
- Step II: Calendar- and outlier adjustment with **RegARIMA**
- Step III: Adjust intra-monthly seasonality with **STL**
- Step IV: Adjust intra-annual seasonality with **STL**

Exercise

Seasonal adjustment with `dsa()`:

- Use respective data from previous exercise
- Default settings
- What do you see?

**Hint: Use `library(xts)` before starting*

**Hint: Use `output()` to generate html with key results*

Seasonal adjustment with DSA (I/II)

```
library(xts)
library(dsa)

series <- xts(data, time)

results <- dsa(series,
               Log = T,                # Add vs mult
               s.window1 = 53,         # Seasonal filter for S7
               s.window2 = NULL,       # Seasonal filter for S31
               s.window3 = 13,         # Seasonal filter for S365
               fourier_number = 24)    # Trigonometric terms
```

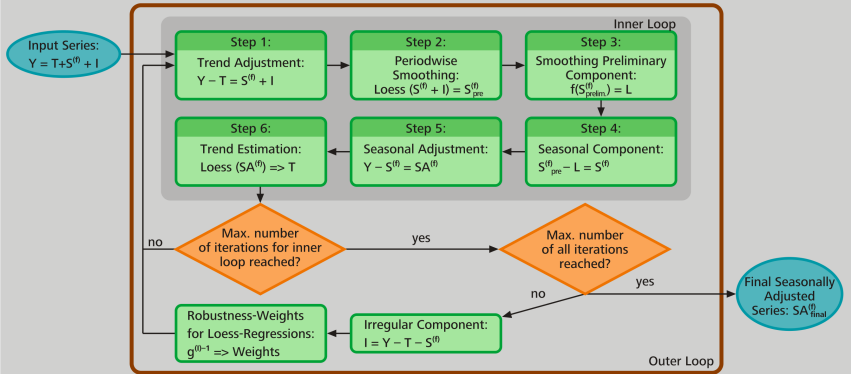
Seasonal adjustment with DSA (II/II)

```
plot(results)                                # Plot DSA results  
  
output(results, path="Path/folder") # Create HTML-output
```

■ DSA output example

[Link to DSAexample.pdf](#)

STL Procedure



Exercise

Determine STL parameters

- Change values for s.window1
- Change values for s.window2
- Change values for s.window3

**Hint: Use output() to generate html with key results*

Exercise

Group

- Group 1: Chile - Retail Sales
- Group 2: India - Electricity
- Group 3: Sri Lanka - ASPI
- Group 4: Korea - Rail Passengers
- Group 5: Colombia - TRM

Example

Electricity consumption: using all calendar regressors

[Link to dsa-vignette.html](#)

Overview of methods

- X-11 extensions
- SEATS variants
- Structural time series models
- TBATS
- STR
- DSA framework

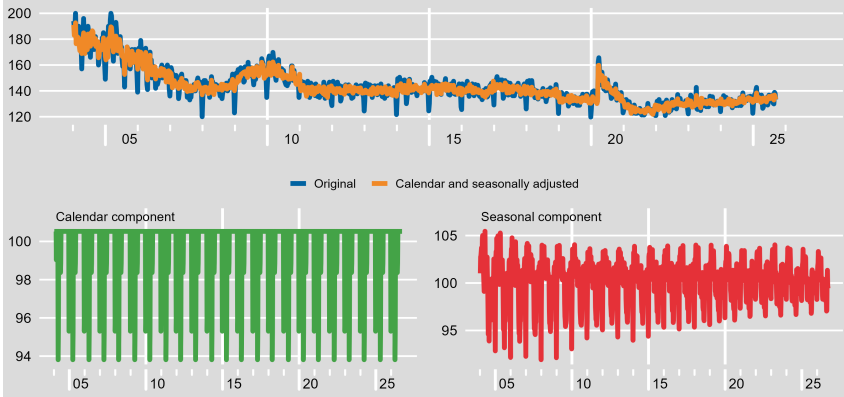
RegARIMA + "Seats"

- RegARIMA: $(p\ d\ q)(P\ D\ Q)_{52} \rightsquigarrow$ Calendar adjustment
- "Seats": Fractional airline model based decomposition $\rightsquigarrow$ Seasonal adjustment

Google Trends

Seasonal adjustment of Google Trends

Search term: Arbeitslosigkeit [Unemployment]



Source: Google Trends, own calculations
Deutsche Bundesbank

Key notions (I/III)

Basic characteristics

Notion	Remark
Many observations	Higher frequency data contain many observations which can be a challenge for algorithms and users.
Short series	Many series contain few months or years of observations.
Temporal aggregation	Restrictions regarding temporal aggregation of adjusted series depending on data type.
Non-isochronicity	Number of observations per periodic cycle is not always the same for all cycles, e.g. number of weeks per year.
Non-equidistance	Some series have non-equally spaced observations, e.g. banking-daily.
Date and time conventions	Conventions regarding the start of the week or year numbering.

Key notions (II/III)

Periodic and calendar effects

Notion	Remark
Multiple periodic effects	Daily time series usually contain day-of-the-week and day-of-the-year effects.
Uncommon periodic effects	Other periodic effects, e.g. week-of-the-month.
Breaks in periodic effects	Periodic effects may change rapidly.
Cross-seasonality	The periodic and calendar effects can be interdependent.
Autocorrelational seasonality	The seasonal impact of adjacent observations may be highly dependent.
Multilevel periodic effects	Seasonal structure may be hierarchical.
Series-specific calendar effects	Regressor construction can more easily be tailored to the series.
Uncommon calendar effects	Other calendar effects, e.g. Daylight Saving Time.
Bridge days	Bridge days may have a traceable effect.

Key notions (III/III)

Outliers and missing values

Notion	Remark
Missing values	Due to data availability, some series contain missing values, often at the end of the series.
Unreliable data delivery	Data producers do not necessarily have an obligation to deliver data or provide additional information.
Higher volatility	The volatility of the time series usually decreases with a higher temporal aggregation.
Heteroskedasticity	The volatility may change over time, e.g. during crises.
Non-traditional outlier patterns	Observed outlier pattern can be different from lower frequency data, e.g. slower rate of decay in TC outlier.

Summary (I/II)

- Multiple seasonalities, e.g.

$$Y_t = T_t + S_t^{(7)} + S_t^{(31)} + S_t^{(365)} + C_t + I_t$$

- Non-integer seasonalities
- More data $\rightsquigarrow$ more information to process

Summary (II/II)

- Higher frequency time series introduce **new challenges** to seasonal adjustment
- Outlook: dsa2 is coming, download here:
<https://github.com/bbkrd/dsa2>
 - You also need packages from here: <https://github.com/rjdverse>

References: Methodology (I/II)

-  Cleveland, R. B., Cleveland, W. S., McRae, J. E. & Terpenning, I. (1990). STL: A Seasonal-Trend Decomposition Procedure Based on Loess. *Journal of Official Statistics*, 6 (1), 3-73.
-  Cleveland, W. S. & S. J. Devlin (1988). Locally Weighted Regression: An Approach to Regression Analysis by Local Fitting. *Journal of the American statistical association*, 83 (403), 596-610.
-  Deutsche Bundesbank (2020). Seasonal Adjustment of the Daily Truck Toll Mileage Index. Methodology Report by the Seasonal Adjustment Section.
-  Eurostat (2015). ESS Guidelines on Seasonal Adjustment. 2015 edition. Publications Office of the European Union, Luxembourg.

References: Methodology (II/II)

-  Ladiray, D., Palate, J., Mazzi, G. L., & Proietti, T. (2018). Seasonal Adjustment of Daily and Weekly Data. In: Handbook on Seasonal Adjustment, G. L. Mazzi, D. Ladiray & D. A. Rieser (eds.). Luxembourg: Publications Office of the European Union, 757–783.
-  Ollech, D. (2021). Seasonal Adjustment of Daily Data. Journal of Time Series Econometrics, 13 (2).
-  Ollech, D. (2022). Economic Analysis using Higher-Frequency Time Series: Challenges for Seasonal Adjustment. Empirical Economics.
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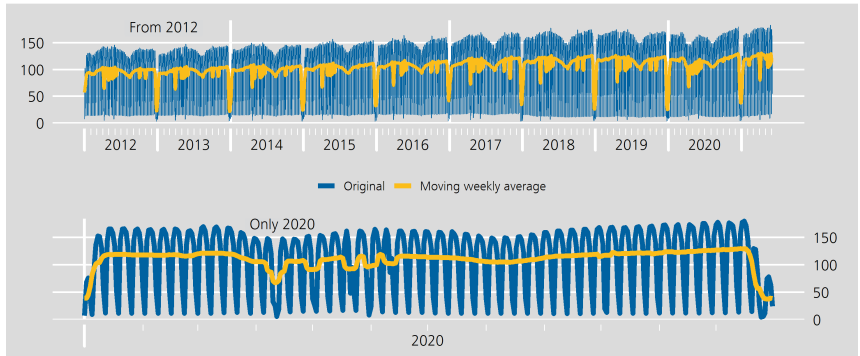
References: Economic indicators

-  Eraslan, S. & Goetz, T. (2020). Weekly activity index for the German economy. Available from <https://www.bundesbank.de/wai>.
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-  Lewis, D. J., K. Mertens, and J. H. Stock (2020). Weekly Economic Index. URL <https://www.newyorkfed.org/research/policy/weekly-economic-index>.

German truck toll mileage index (I/V)

German truck toll mileage index

2015=100



Source: Federal Office for Freight Transport, own calculations
Deutsche Bundesbank

German truck toll mileage index (II/V)

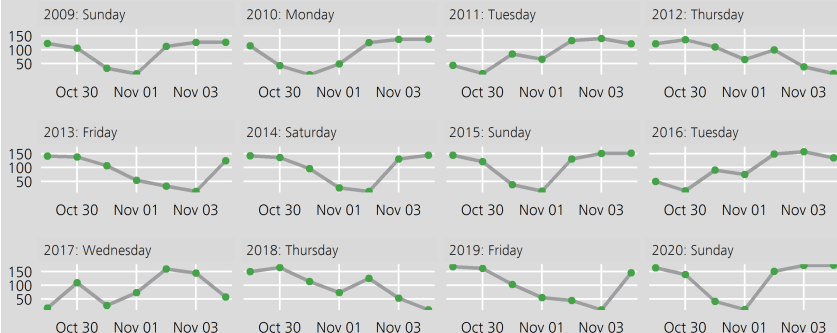
Stylised facts

- Many observations
- Multiple periodic effects
- Break in periodic pattern
- Cross-seasonality
- Bridge days
- Higher volatility than monthly series

German truck toll mileage index (III/V)

German truck toll mileage index

2015=100, days around All Saints' Day (November 1)



Remark: Subplot headline indicates weekday of All Saint's Day.

Source: Federal Office for Freight Transport, own calculations.
Deutsche Bundesbank

German truck toll mileage index (IV/V)

Moving holiday and cross-seasonal effects

	Estimate	S.E.		Estimate	S.E.
Carnival Monday	-8.6	1.6	NH ¹ (Mon)	-92.3	2.3
Mardi Gras	-5.0	1.6	NH ¹ (Tue)	-105.9	2.4
Holy Thursday	-20.0	1.6	NH ¹ (Wed, to 2015)	-90.2	3.1
Good Friday	-99.6	1.9	NH ¹ (Wed, from 2015)	-118.7	2.7
Holy Saturday	-17.3	1.9	NH ¹ (Thu)	-106.4	2.1
Easter Sunday	-11.1	2.0	NH ¹ (Fri)	-74.5	1.9
Easter Monday	-116.8	1.9	NH ¹ (Sat)	-21.2	2.1
Easter Mon. (t+1)	-20.4	1.6	Xmas Period (Mon)	-31.7	1.8
Ascension (t-1)	-13.6	1.6	Xmas Period (Tue)	-42.6	1.8
Ascension (to 2015)	-108.9	2.3	Xmas Period (Wed)	-42.0	1.8
Ascension (from 2016)	-122.1	2.5	Xmas Period (Thu)	-41.7	1.8
Ascension (t+1)	-17.4	1.7	Xmas Period (Fri)	-29.0	1.8
Corpus Chr. (t-1, to 2015)	-7.3	2.0	3d before Xmas (Sun)	120.1	18.9
Corpus Christi (to 2015)	-73.5	2.3	Xmas Eve (Sat)	11.4	4.4
Corpus Christi (from 2016)	-77.6	2.3	Xmas Eve (Sun)	42.1	5.2
Corpus Chr. (t+1)	-7.3	1.6	Xmas Day (Sat)	22.0	5.9
Pentecost (t-1)	4.2	1.3	Xmas Day (Sun)	59.2	4.3
Pentecost (to 2015)	-107.2	1.9	Boxing Day (Sat)	18.7	3.2
Pentecost (from 2016)	-119.3	2.3	Boxing Day (Sun)	33.2	5.9
Pentecost (t+1)	-13.4	1.6	10d post Dec 26 (Sat)	31.8	2.0
German Unity (bridge)	-2.4	2.9	10d post Dec 26 (Sun)	14.8	1.9
Labour Day (bridge)	-17.7	2.6	All Hallows (bridge)	-20.6	2.4

German truck toll mileage index (IV/V)

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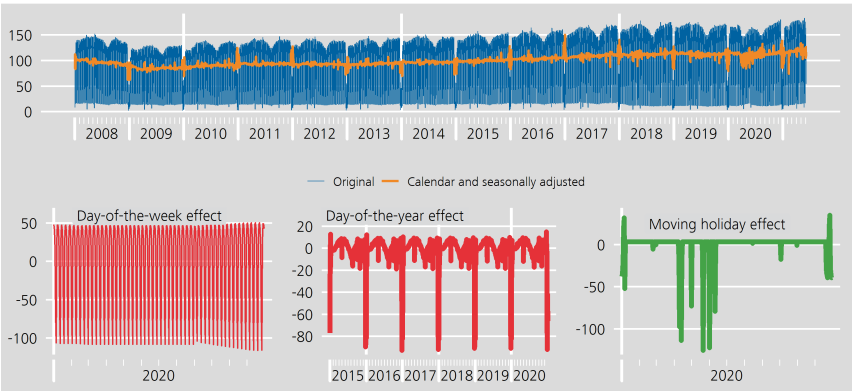
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German truck toll mileage index (V/V)

Seasonal adjustment of German truck toll mileage index

2015=100

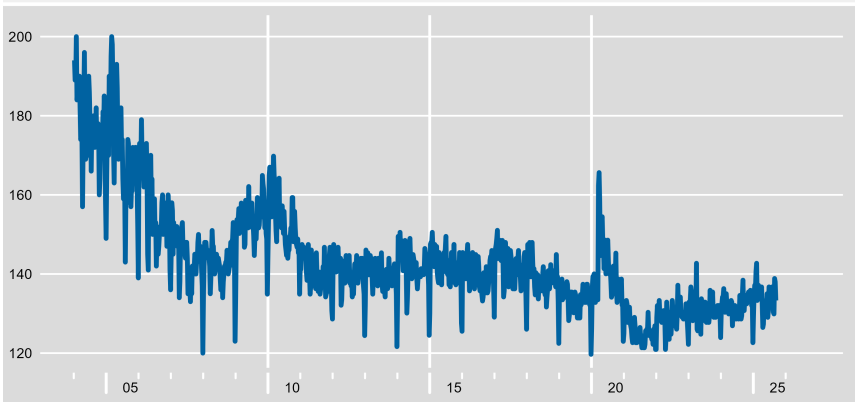


Source: Federal Office for Freight Transport, own calculations
Deutsche Bundesbank

Google Trends (I/III)

Seasonal adjustment of Google Trends

Search term: Arbeitslosigkeit [Unemployment]



Source: Google Trends, own calculations
Deutsche Bundesbank

- Non-isochronicity
- Heteroskedasticity
- Unreliable data delivery

Google Trends (III/III)

ARMA coefficients and moving holiday effects (in percent)

Estimate			Estimate		
S.E.			S.E.		
MA	−0.87	0.02	SMA	−0.80	0.03
Good Friday	−6.5	1.1	Christmas Eve	−10.0	1.1
Easter	−4.8	1.1	New Year's Eve	−8.9	1.4